PIYUSH SONI

AI Engineer | Applied AI & LLM Application Developer | Python Backend | AI Trainer

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PROFESSIONAL SUMMARY

AI Engineer who turns rough ideas into shipped, usable AI products — reasoning about what the product needs, designing the flow, building it end to end, and refining it based on real-world performance. Production experience at Sara Analytics building LLM-powered applications, RAG pipelines, multi-agent orchestration, and document-intelligence tools using the Anthropic Claude, OpenAI, Google AI Studio (Gemini), and AWS Bedrock APIs with LangChain and vector databases (ChromaDB, FAISS). Ships features as FastAPI/Flask REST services, containerized with Docker and deployed through CI/CD, with Streamlit, Taipy, HTML/CSS, and AI-assisted React front ends, built using AI-assisted coding tools (Google Antigravity, VS Code, CLI) — while reviewing and understanding the code generated. Currently a Trainer at SPSU teaching Deep Learning and Reinforcement Learning. Came into tech from a non-technical background, which makes explaining complex problems in plain language a genuine strength; brings a business mindset focused on turning unstructured data into structured, valuable products.

CORE COMPETENCIES

LLM Application Development • Prompt Engineering & Structured Outputs • Tool / Function Calling • RAG & Vector Databases • Multi-Agent Orchestration • AI Workflow Design (Context Handling, Guardrails, Fallbacks) • API Integration • AI-Assisted Frontend Development (React) • System Architecture • Docker & CI/CD • AI Testing & Iteration • Debugging • Technical Communication

TECHNICAL SKILLS

Languages: Python (OOP), SQL, HTML/CSS, JavaScript & React (AI-assisted development)

LLM APIs & Platforms: Anthropic Claude API, OpenAI API, Google AI Studio (Gemini), AWS Bedrock, Hugging Face Transformers, Ollama, LM Studio

AI Engineering: Prompt engineering, structured JSON outputs, tool/function calling, context handling, response validation, LLM orchestration, AI agents & multi-agent systems, model fine-tuning, managing AI cost, latency & limitations

RAG & Vector Databases: LangChain, ChromaDB, FAISS, embeddings, semantic retrieval pipelines

Backend & APIs: FastAPI, Flask, REST API design, route & endpoint design, JSON, JWT authentication, role-based access control, third-party API integration

Frontend & App Frameworks: HTML, CSS, Streamlit, Taipy, React (AI-assisted); basic UI/UX principles — clarity, hierarchy, consistency

AI-Assisted Coding & Tools: Google Antigravity, VS Code, CLI-based development, Git & GitHub (branching, commits, pull requests)

Deployment & DevOps: Docker, CI/CD pipelines, PM2, systemd service files, Nginx, Ubuntu Linux, AWS, Vercel

Data & Machine Learning: Databricks, PostgreSQL, Supabase, MySQL, SQL optimization, EDA, data preprocessing, feature engineering, regression, clustering, PCA, Deep Learning, Reinforcement Learning, Power BI

PROFESSIONAL EXPERIENCE

Trainer — Deep Learning & Reinforcement Learning **Sep 2026 – Present**

Sir Padampat Singhania University (SPSU) | Udaipur, Rajasthan

- Teach Deep Learning and Reinforcement Learning to university students, covering neural network fundamentals, model training and evaluation, and core RL concepts such as agents, environments, rewards, and policies.
- Translate complex AI concepts into plain-language explanations and hands-on examples, drawing on real production projects built in industry.
- Guide students through practical coding exercises and projects, debugging their models and code and connecting theory to real-world AI applications.

AI/ML Developer **Sep 2025 – Jul 2026**

Sara Analytics Private Limited | Jaipur, Rajasthan | Experience Letter | Projects on Company Platform

- Designed and shipped LLM-powered applications end to end — from business requirements and workflow design through backend APIs, frontend, deployment, and iteration — using the Anthropic Claude, OpenAI, Google AI Studio, and AWS Bedrock APIs.
- Architected PeopleOS, a multi-agent HR platform with a RAG-based orchestration layer that unifies recruitment, onboarding, attendance, and employee-information agents behind a single chatbot, so users complete every task in one place instead of opening separate tools.
- Built RAG pipelines and enterprise knowledge-retrieval chatbots with LangChain, ChromaDB, Ollama, and LM Studio, designing context handling and retrieval to keep responses grounded in company data.
- Built Lease Compliance Analysis, a prompt-engineered LLM workflow producing structured outputs: extracts key clauses from lease and contract documents, checks them against regulatory and organizational rules, flags high-risk terms, and auto-generates compliance reports.

- Built GL Analyzer, an automated financial data tool that ingests records across accounting formats (QBO, MIP, QBD), standardizes them, and generates structured reports.
- Built TalentScout, an end-to-end recruitment pipeline integrating the LinkedIn Developer API and Zimyo API to automate resume screening, candidate-to-role matching, and LLM-based assessment generation.
- Built user-facing interfaces for AI agents and internal tools with HTML/CSS and AI-assisted React development, plus Streamlit and Taipy apps, delivering clean, usable products without a dedicated frontend team.
- Developed FastAPI and Flask REST services for model serving and workflow automation, designing routes and endpoints with JWT authentication and role-based access control.
- Containerized services with Docker, set up CI/CD pipelines, and deployed to Ubuntu servers using PM2, systemd service files, and Nginx; managed PostgreSQL databases.
- Tested and iterated on AI behavior — validating model responses, catching edge cases, and refining prompts for consistency — and regularly debugged production issues across backend, data, and LLM layers.
- Built production Databricks workflows for data ingestion, transformation, and validation; wrote and optimized SQL/PostgreSQL queries; implemented ML-based anomaly detection; collaborated via Git and GitHub.

Teaching Assistant (Intern) **Mar 2025 – Aug 2025**
AlmaBetter | Remote | Experience Details

- Mentored learners in Python, SQL, Statistics, Machine Learning, Excel, and Power BI; ran doubt-resolution sessions and debugged student projects.

Chatbot Developer / AI **Dec 2023 – Feb 2025**
Adclickr (Advertising & Marketing Agency) | Jaipur, Rajasthan | Experience Certificate

- Designed and built rule-based and NLP-driven chatbots in Python for website and messaging channels, automating customer support and lead-response workflows.
- Developed intent-recognition and conversation-flow logic, improving response accuracy and reducing average query-resolution time.
- Integrated chatbot backends with REST APIs and databases for real-time data, and iterated on flows using user feedback and conversation analytics with the product team.

KEY PROJECTS

Bank Statement & Invoice Extractor | AWS Bedrock, OCR, LLM Extraction

- Built an OCR + LLM pipeline that converts scanned and image-based bank statements and invoices into clean, structured data for downstream reporting and validation.

PeopleOS — Multi-Agent Orchestration Architecture | LangChain, RAG, AI Agents

- Designed independent agents (Time Tracker, People Insights, TalentScout, OnboardMate), each owning its own data domain, with an orchestration layer routing requests and coordinating hand-offs between agents.

Stockroom — Inventory Management System | Supabase, Vercel | Live Project

- Built and deployed an inventory management web application using Supabase as the backend and database, hosted on Vercel and accessible through a live login page.

Machine Learning Projects | scikit-learn, EDA, Power BI

- Yes Bank Stock Price Prediction: regression models (Linear, Lasso, Ridge, ElasticNet, KNN) with GridSearchCV tuning, evaluated on R^2, MAE, and RMSE.
- Customer Segmentation: EDA, RFM analysis, feature engineering, PCA, and K-Means clustering to define customer segments for targeted marketing.
- E-commerce Sales Dashboard: ETL workflows in Pentaho into a SQL Server warehouse, visualized in an interactive Power BI dashboard.

EDUCATION

Master of Computer Applications (MCA) — IGNOU Feb 2025 – Present
Diploma in Data Science & Analytics (Full Stack Data Science) — AlmaBetter | Certificate 2024 – 2025
Advanced Certification in Full Stack Data Science & AI — IIT Guwahati | Certificate 2023 – 2024
Bachelor of Arts (BA) — University of Rajasthan | Certificate 2017 – 2020
Senior Secondary (Class XII) | Marksheet
Secondary (Class X) | Marksheet

CERTIFICATIONS

- Generative AI & Prompt Engineering — Udemy | Certificate
- Full Stack Data Science & AI — AlmaBetter | Certificate
- Advanced Certification in Full Stack Data Science & AI — IIT Guwahati | Certificate
- Python (Basic) — HackerRank | Certificate